Technology Stack

Date	30-Jun-2026
Team ID	xxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

The OptiCrop application uses a modern and lightweight technology stack to develop an intelligent crop recommendation system. The selected technologies support efficient machine learning model integration, responsive user interface development, reliable backend processing, and seamless deployment. They also simplify maintenance and future enhancements.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side**	HTML5, CSS3, Bootstrap 5, JavaScript, Jinja2 Templates	These technologies are used to develop a responsive and user-friendly interface that enables farmers to enter soil and weather information and easily view crop recommendations on different devices.

2	**Backend / Server-Side**	Python 3.11, Flask 2.2.2	Flask is a lightweight web framework that efficiently handles user requests, integrates the trained machine learning model, and returns prediction results with minimal delay.
3	**Database / Data Storage**	Crop Recommendation Dataset (CSV), Joblib/Pickle (.pkl) File	The CSV dataset is used for model training, while the serialized model file stores the trained Random Forest model for fast prediction without retraining.
4	**Cloud / Hosting / Deployment**	Render, Procfile	Render provides an easy cloud deployment platform with GitHub integration, automatic builds, and reliable hosting for the Flask application.

5	**Version Control & CI/CD**	Git, GitHub	Git and GitHub are used to manage source code, maintain version history, support collaborative development, and simplify project management.
6	**Third-Party APIs / Other Tools**	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook	These libraries support data preprocessing, exploratory data analysis, visualization, machine learning model development, training, testing, and evaluation for the OptiCrop system.